# SIGHT Project - Meeting 27

**Project Name:** Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

**Project Number:** WSIC26-250206-009

**Date:** October 16, 2026

**Time:** 11:00 AM to 12:11 PM EDT

**Location:** Zoom

**Meeting Focus:** Recap of Meeting 26 action items, BWC deliverables and the Q6 deck, the October 14 AM Hub visit, the OSC 2027 session, the digital twin, the cost-benefit study, the Bridgeport mill SOP, computer vision progress, and the risk log.

## Attendees

- **Miami University (MU):** Mohammad Mayyas (Co-PI, Meeting Chair), Jay Shan (Co-PI), Ibrahim Yousif (Associate Research Scientist), Michael Wise (GRA), Austin Hamilton (GRA/SIGHT AR Lead), Lindsay Stevenson (Student Research Assistant), Leo Lyublinski (Student Assistant), Wissam Nusair (Student Research Assistant), Mo Farrag (Project Manager)
- **Discovery Center:** Kristen Morio (Evaluator), Abby Helsinger (Senior Research Associate/Evaluator)
- **External Consultant:** Lora Cavuoto (University at Buffalo)
- **MaxByte:** Ramshankar C S (CEO)
- **Industrious Group Inc.:** Jeff Malek (Corporate Director of EHS)

## 1. Introductory Remarks and Recap / Approval of Minutes

- **Welcome & Recap:** Mohammad Mayyas chaired the meeting on behalf of Fadel Megahed. Mohammad reviewed the Q5 report, the Farmer School session, the SIGHT-VR v1.0 release, and the first SOP-guided mill run.
- **Approval of Minutes:** The Meeting 26 minutes were available through the deck. The team raised no corrections.
- **Status of Meeting 26 Action Items:**
    - Q5 report draft and Farmer School session: done.
    - Onboarding for first-time AR headset users: in progress. Michael gave Austin feedback from first-time users. Austin focused on the manual mill during the past two weeks.

- Computer vision model for Austin: Ibrahim reported progress and showed it later in the meeting. Ibrahim explained that the model must match the CAD model so that it detects only the equipment. Mohammad recorded this item as complete.
- Safety-zone circle: Austin checked the circle after re-anchoring and reported that it stayed centered.
- AR evaluation data with the VR backend: open on the AR side. Arthur Carvalho will report the VR side.
- IRB amendment: open. Lora will draft the amendment and send it to Jay. Jay will combine it with a required IRB report.
- Experimental protocol and schedule: open. Lora will attend the Human Factors conference from Monday to Wednesday. The team will offer times on Tuesday, October 20, for a Zoom meeting.

## 2. Administrative Updates

- **BWC Deliverables Status:**
  - The team completed 11 of the 15 project tasks.
  - BWC approved all deliverables due through September 11, 2026.
  - The Q5 report and the September/October meeting documents are pending BWC review.
  - The next items are the Q6 deck (October 19), the quarterly grant meeting (October 23), and the Q5 expenditure report and invoice (October 30).
- **BWC Q6 Presentation:** Mohammad showed the deck. It includes project status, budget, the Makino VR extension, and the SOP work. Team members will send comments to Fadel on October 16, because the deck is due on Monday, October 19.
- **OSC 2027 Session:** BWC accepted the SIGHT session on the use of generative AI to develop training materials. The session is on March 18, 2027, from 10:30 to 11:30 AM, at the Ohio Safety Congress and Expo in Columbus. Mohammad and Ibrahim will attend. Arthur and Jay will check their calendars.
- **AM Hub Visit (October 14):** President Crawford and Senator Moreno visited the AM Hub. The team demonstrated the AR and VR technology. Mohammad reported that President Crawford plans to mention this work in an upcoming address.
- **New Team Members:** Usman Asif joined the team as a student researcher. Usman has about nine years of occupational health and safety experience, in addition to an IT background. Om

Raval is the second hire and is eligible to start. Jay will ask both students to complete the CITI training so that the IRB can include them.

- **Makino Update:** Mohammad shared the D200Z and UX3i VR trainings with Makino, and discussions continue. Mohammad is arranging a visit by a Makino technical manager. Michael is waiting for a date for the visit to Makino.

## 3. Technical Updates: MU

- **Digital Twin (Wissam Nusair):**
  - Wissam showed the digital twin website, which has separate views for managers, engineers, and specialists. These views show lab status, room layouts, telemetry, alerts, and camera feeds.
  - All data in the demonstration was simulated because the machines were off. The site can receive live machine data through MQTT.
  - Ibrahim reported that the connected machines already send data, and that this task improves the user interface. The head count in each lab is simulated for now. It needs input from machine vision and ID access records.
- **Cost-Benefit Analysis (Leo Lyublinski):**
  - Leo showed the current paper, formatted for the 1,500-word limit of Manufacturing Letters. The paper uses the benefit-cost ratio, net present value, and payback period.
  - The analysis uses four separate benefit streams: training, safety, quality, and downtime. It solves for the injury reduction that an investment needs to break even, because proving an injury reduction would need a multi-year, multi-site test.
  - Mohammad asked how the model relates safety to production efficiency. Leo explained that the equations prevent double counting across the benefit streams.
  - Next Steps: Leo is working on the second draft and the case study design. Leo expects results in about four weeks. The study needs payroll data and OSHA logs from an industry partner (see Section 4).
- **Bridgeport Mill SOP and Haas Certification (Michael Wise):**
  - The Bridgeport mill SOP is complete and Austin has it. The work took four weeks and produced a 37-page SOP from a 108-page manual.
  - Michael considers the mill the highest-risk machine in the project. The SOP tells users to verify that the machine is off before they touch the spindle area. Michael also designed a flexible 3D-printed TPU cover for the cutting flutes.

- Michael completed 13 of 16 mill modules and 4 of 15 lathe modules in the Haas certification outside project hours. Michael proposed that the SOPs explain feed hold. Michael also proposed that the VR and AR content adjust rapid traverse and not feed rate.
- Ibrahim asked how a novice can recognize an abnormal machine sound, and how the system can convert that sound into an action. Michael will find a video example. The team will continue this discussion in a small group.
- Mohammad asked if users with no technical background can follow the SOP. So far, Jay tested it and Austin tried it. Michael offered to test it with other volunteers.

- **Computer Vision and Dissemination (Ibrahim Yousif):**
  - The public RoboFlow dataset grew from about 5,600 to 10,000 annotated images under Miami University. It includes the CNC lathe, the mill, the cobot, people, and the control pendant.
  - The human and robot detection model reports 99% accuracy. Ibrahim noted that this figure has little meaning until the team tests the model in extreme conditions, such as different people and clothing.
  - Next Steps: Ibrahim will add depth so that the system can calculate the distance between the person and the robot. Mohammad asked Ibrahim to convert pixels to physical distance, based on the camera setup, to support the safety zones.
  - Ibrahim showed early work on a vision model that uses CAD data with no real machine images. Ibrahim reported about 50% accuracy from CAD data alone.
  - Ibrahim asked the team to include machine interlocks and permissives in the AR, so that the system can recognize an unauthorized action. Ibrahim also showed a phone-based AR view of the machines, and Mohammad asked how the view verifies each step of a G-code program.
- **Farmer School Presentation (Arthur Carvalho):** Arthur did not attend. Mohammad reported that Arthur presented SIGHT on October 9. Jay reported that the dean and the associate dean attended, and that the team received positive feedback.

## 4. Industry Partner Collaboration

- **MaxByte Update:** MaxByte did not present at this meeting. Ibrahim reported that MaxByte sent an email about this. Ramshankar C S left for another call before this item. Mohammad will contact Ramshankar about the Unity license and the open items from Meeting 26.

- **Industrious Group Inc.:** Jeff Malek offered to share preliminary data for the cost-benefit model, which can include financial data and BWC payroll reports. Jeff asked Leo to make direct contact. Jeff left for another call before the end of the meeting.

## 5. Risk Mitigation and Log Updates

- **Hardware Obsolescence (Magic Leap 2):** The backup headset is available. This risk is mitigated.
- **IRB Amendment:** This risk remains open. The team needs the amendment before it records sessions.
- **AR Operation:** Austin identified that only one person could run the AR system. Austin trained Michael, and Michael confirmed the ability to configure the system for the CNC lathe and the cobot. Michael is now the backup.
- **SOP Verification:** The team wants to verify that users with no technical background can follow the SOPs before the team converts them to AR instructions.

## 6. Further Discussion and Wrap-Up

- **Discovery Center Request:** Kristen Morio asked the team to tell the evaluation team when the IRB approves the amendment and when testing is scheduled. Kristen serves on the IRB committee and offered to help. Kristen will attend the follow-up meeting.
- **Follow-Up Meeting:** Michael or Ibrahim will schedule a meeting with Lora, Jay, Kristen, and Fadel, if Fadel is available. The meeting will review the work instructions and any additional IRB amendment needs for VR and AR.
- **Next Meeting:** Mohammad stated that the next bi-monthly meeting is on November 13, 2026.

## Action Items

| Task | Responsible | Deadline |
|---|---|---|
| Send comments on the Q6 deck to Fadel Megahed | All team members | October 16, 2026 |
| Submit the Q6 deck to BWC | Fadel Megahed | October 19, 2026 |

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| Check calendars for the OSC 2027 session on March 18, 2027 | Arthur Carvalho / Jay Shan | October 19, 2026 |
| Schedule the protocol and work instruction meeting with Lora, Jay, Kristen, and Fadel | Michael Wise / Ibrahim Yousif | October 20, 2026 |
| Draft the IRB amendment and send it to Jay | Lora Cavuoto | ASAP |
| Combine the amendment with the IRB report and add the new hires | Jay Shan | ASAP |
| Ask Usman Asif and Om Raval to complete the CITI training | Jay Shan | ASAP |
| Submit the Q5 expenditure report and Q5 invoice | Fadel Megahed | October 30, 2026 |
| Complete onboarding for first-time AR headset users | Austin Hamilton / Michael Wise | Before the Cleveland visit |
| Store AR evaluation data with the VR backend and create one participant ID across VR and AR | Austin Hamilton / Arthur Carvalho | Before pre-pilots |
| Contact Jeff Malek about preliminary data for the cost-benefit model | Leo Lyublinski | Next update |
| Complete the second draft and the case study | Leo Lyublinski | About four weeks |
| Add feed hold to the SOPs and review the feed rate and rapid traverse instructions in VR and AR | Michael Wise | Next update |
| Find a video example of abnormal machine sounds | Michael Wise | Next update |
| Test the Bridgeport mill SOP with volunteers who have no technical background | Michael Wise | Before AR testing |

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| Add depth, convert pixels to physical distance, and test the vision model in extreme conditions | Ibrahim Yousif | Next update |
| Contact Ramshankar C S about the Unity license and the open Meeting 26 items | Mohammad Mayyas | ASAP |
| Arrange the visit by a Makino technical manager | Mohammad Mayyas | ASAP |
| Confirm the date for the visit to Makino | Michael Wise | When Makino responds |
| Add Usman Asif to the meeting invitation | Mohammad Mayyas | Next meeting |
| Tell the Discovery Center when the IRB approves the amendment and when testing is scheduled | Jay Shan / Mohammad Mayyas | When confirmed |